

Math 571 - Homework 5

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Problem 5.1 (R:5.8). Suppose f' is continuous on $[a, b]$ and $\varepsilon > 0$. Show that there is $\delta > 0$ so that for all t such that $0 < |t - x| < \delta$ and all $a \leq x \leq b$

$$\left| \frac{f(t) - f(x)}{t - x} - f'(x) \right| < \varepsilon$$

This could be stated as f' is *uniform continuity* on $[a, b]$ provided f' is continuous on $[a, b]$. Does this hold for vector-valued functions?

Problem 5.2 (R:5.16). Suppose f is twice differentiable on $(0, \infty)$ and f'' is bounded on $(0, \infty)$, and $f(x) \rightarrow 0$ as $x \rightarrow \infty$. Show that $f'(x) \rightarrow 0$ as $x \rightarrow \infty$.

Problem 5.3 (R:5.22). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable on (a, b) and continuous on $[a, b]$. A point x is a **fixed point** of f iff $f(x) = x$.

- (a) Show that if $f'(t) \neq 1$ for all $t \in (a, b)$, then f can have at most one fixed point.
- (b) Show that $f(t) = t + (1 + e^t)^{-1}$ satisfies $|f'(t)| < 1$ and f has no fixed points.
- (c) Show that if there is $A < 1$ so that $|f'(t)| \leq A$ for all $t \in \mathbb{R}$, then f has a fixed point and moreover given any $x_0 \in \mathbb{R}$ and taking $x_{n+1} = f(x_n)$ it turns out that $x_n \rightarrow x$ and $f(x) = x$ is the unique fixed point of f .
- (d) Show that the process described in (c) can be visualized by the zig-zag path

$$(x_1, x_2) \rightarrow (x_2, x_2) \rightarrow (x_2, x_3) \rightarrow (x_3, x_3) \rightarrow (x_3, x_4) \rightarrow \cdots$$

Problem 5.4 (R:5.24). The process described in part (c) of Exercise 22 can of course also be applied to functions that map $(0, \infty)$ to $(0, \infty)$. Fix some $\alpha > 1$, and put

$$f(x) = \frac{1}{2} \left(x + \frac{\alpha}{x} \right), \quad g(x) = \frac{\alpha + x}{1 + x}.$$

Both f and g have $\sqrt{\alpha}$ as their only fixed point in $(0, \infty)$. Try to explain, based on the basis of properties of f and g , why the convergence in Exercise 16, Chap. 3, is so much more rapid than it is in Exercise 17. (Compare f' and g' , draw the zig-zags suggested in Exercise 22.) Do the same when $0 < \alpha < 1$.

Hint: A better hint than that given is to investigate using Taylor's Theorem with remainder with just three terms: $f'(x) = f(\sqrt{\alpha}) + f'(\sqrt{\alpha})(x - \sqrt{\alpha}) + \frac{1}{2}f''(c)(x - \sqrt{\alpha})^2$ for some c between x and $\sqrt{\alpha}$.

Problem 5.5 (B5.2:2 - A fake counterexample to Theorem 5.9.). The conclusion of Theorem 5.9 says, roughly, that if we look at the parametrized curve $(f(t), g(t))$, then for some value of t in (a, b) , the ratio of the x - and y -components of its velocity will coincide with the ratio of the x - and y -components of the total displacement, $(f(b) - f(a), g(b) - g(a))$; i.e., that there is a point where the direction of the curve is parallel to the line segment from $(f(a), g(a))$ to $(f(b), g(b))$.

However, consider the case $f(t) = t^2$, $g(t) = t^3$ on the interval $[-1, 1]$. Sketch the graph, being especially careful near $t = 0$.

- Is there a point where the curve is parallel to the line segment referred to? For what value of t does the equation of Theorem 5.9 hold?
- How does the curve look at the corresponding point? (That is why I wrote "roughly" in the first sentence of this exercise.)
- Under what **additional condition** on the curve $r(t) = (f(t), g(t))$ for $t \in [a, b]$ can we remove the word roughly and exactly that at some $t_0 \in (a, b)$, $r'(t_0) = (f'(t_0), g'(t_0))$ is parallel to $(f(b) - f(a), g(b) - g(a))$?

Problem 5.6 (R:5.26). Suppose $f(x)$ is differentiable on $[a, b]$, $f(a) = 0$, and there is a fixed A such that $|f'(x)| \leq A|f(x)|$ for all x in $[a, b]$. Show that $f(x) = 0$ on $[a, b]$.

The next problem makes use of the one you just did.

Problem 5.7 (R:5:27). Let $\phi : [a, b] \times [\alpha, \beta] \rightarrow \mathbb{R}$. A *solution to the initial-value problem* (IVP)

$$y' = \phi(x, y), \quad y(a) = c \text{ for } a \leq c \leq b$$

is a function $f : [a, b] \rightarrow [\alpha, \beta]$ satisfying

$$f(a) = c, \quad f'(x) = \phi(x, f(x)) \text{ for all } a \leq x \leq b$$

Show that if there is a constant $A \geq 0$ so that

$$|\phi(x, y_1) - \phi(x, y_2)| \leq A|y_1 - y_2| \text{ for all } x \in [a, b] \text{ and } y_1, y_2 \in [\alpha, \beta],$$

then there is at most one solution to any such IVP.

Problem 5.8 (B:5.6:1 - A function whose Taylor polynomials converge to the wrong result). Here is a well-known example concerning Taylor series, which Rudin doesn't give till Chapter 8 (where it is Exercise R:8.1), because it uses properties of the exponential function, which he develops there. Since we don't reach that chapter in this course, let us assume for this exercise a few basic properties of that function: That e^x is everywhere nonzero, that $e^{-x} = 1/e^x$, that e^x is differentiable, with derivative equal to itself, and that $e^x \rightarrow +\infty$ as $x \rightarrow +\infty$. We begin by deducing a few more facts from these.

- Show that for every polynomial $p(x)$, as $x \rightarrow +\infty$ we have $p(x)e^{-x} \rightarrow 0$. (Hint: Use L'Hospital's rule and induction on the degree of p .)

- (b) Sketch an argument showing from this that for every polynomial $p(x)$, $\lim_{x \rightarrow 0} p(x^{-1})e^{-x^{-2}} = 0$. (I say "sketch" because, although Rudin proves in Theorem 4.7 that for continuous functions f and g with appropriate domains and codomains, the composite function $g \circ f$ is also continuous, he does not develop general results on *limits* of composite functions; in particular limits at infinity; and although it is not too difficult to prove such results, it would be time-consuming to have you do so here. So simply sketch how these functions behave, assuming that in this example, limits of composites behave as one would expect. Note that " $x \rightarrow 0$ " involves values both above and below 0.) We can now give the example. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = e^{-x^2}$ if $x \neq 0$, and $f(0) = 0$.
- (c) Show that on $\mathbb{R} \setminus \{0\}$, f has derivatives $f^{(n)}$ for all nonnegative integers n , and that for each n , $f^{(n)}(x)$ has the form $p_n(x^{-1})e^{-x^{-2}}$ for some polynomial $p_n(x)$.
- (d) Show that for every nonnegative integer n , $f^{(n)}(0)$ is also defined, and equals 0. Now compute the Taylor polynomials shown in display (23) on p.110 for $a = 0$. Show that these converge as $n \rightarrow \infty$, but that the limit is not the original function $f(t)$.
- (e) As a variant of the above, suppose we define a function g by $g(x) = e^{-x^{-2}}$ if $x > 0$, and $g(0) = 0$ if $x \leq 0$. Show that g is also infinitely differentiable, and is given by its Taylor series for all negative x , but not for any positive x .

Problem 5.9. For our finale, here are a couple of background things to look at [Wikipedia](#) and a Quanta article [The Jagged, Monstrous Function That Broke Calculus](#).

Our final problem will build off of the last problem in Homework 4. Recall from there that $C([0, 1]) = \{f : [0, 1] \rightarrow \mathbb{R} \mid f \text{ is continuous}\}$ is a complete metric space under the **uniform metric** or ∞ -norm. To review, $\|f\|_\infty = \sup\{f(x) \mid x \in [0, 1]\}$ and by the extreme value theorem, this could be replaced with $\|f\|_\infty = \max f([0, 1])$. We will use the Baire Category Theorem in $C([0, 1])$ to show that a large set (co-meager set) of continuous functions are nowhere differentiable.

The goal here is not to look at a specific example of a continuous nowhere differentiable function, but rather to show that, in some sense, **almost all** continuous functions are nowhere differentiable.

Define

$$D = \{f \in C([0, 1]) \mid f \text{ is differentiable at at least one point } x_0 \in [0, 1]\}$$

Define

$$E_n = \{x \mid (\exists x \in [0, 1])(\forall y \in [0, 1])(|f(x) - f(y)| < n|x - y|)\}$$

- (a) Suppose $f \in C([0, 1])$ is differentiable at a point $x_0 \in [0, 1]$. Show that there is $n \in \mathbb{N}$ so that $|f(x) - f(x_0)| < n$ for all $x \in [0, 1]$. That is $D \subseteq \bigcup_n E_n$.
- (b) Show that E_n is closed. For this, take $f_i \in E_n$, suppose $f_i \rightarrow f$ uniformly. Show $f \in E_n$.

Hint: Take $x_i \in [0, 1]$ so that $(\forall y \in [0, 1])(|f_i(x_i) - f(y)| \leq n|x_i - y|)$. Since $[0, 1]$ is compact find a subsequence x_{m_i} so that $x_{m_i} \rightarrow x \in [0, 1]$. (sequential compactness).

- (c) Show that $\text{int}(E_n) = \{\}$. That is, E_n is *nowhere dense*, or conversely, E_n^c is dense.

Hint: See this [visual hint \(Desmos\)](#). Fix $\epsilon > 0$, find a piecewise linear $l(x) \in C([0, 1])$ so that $\|f - l\|_\infty$ is small. Let $s(x)$ be a *sawtooth function*, on $[0, 1]$ we have

$$s(x) = \begin{cases} 2x - \frac{1}{2} & \text{if } x \in [0, \frac{1}{2}] \\ -2x + \frac{3}{2} & \text{if } x \in [\frac{1}{2}, 1] \end{cases}$$

Furthermore s is periodic, $s(x + 1) = s(x)$. We can consider $as(bx)$ for b an integer with period $\frac{1}{b}$, or frequency b , and *amplitude* $\frac{a}{2}$. At any point $x_1 \in [0, 1]$, $|as(bx) - as(bx_1)| = 2ab$ for so x with $|x - x_1| < \frac{1}{b}$. So by choosing a small, we get $\|as(bx)\|_\infty = \frac{|a|}{2}$ while at the same time by choosing b large we can ensure that $as(bx) \notin E_n$. Consider now $h(x) = l(x) + as(bx)$ as a point in $C([0, 1])$ close to $f(x)$, but with the property that $\left| \frac{h(x) - h(y)}{x - y} \right|$ can be made arbitrarily large.

- (d) Apply the Baire Category Theorem to see that the set of nowhere differentiable functions is a dense G_δ (**co-meager**) set in $C([0, 1])$.